

AK VisionFlow — Phase 7

Claude Development Prompt
Advanced Optical + Eye Clinic + CRM

Critical context: Phase 1–6 provide the existing operational, ERP, accounting, procurement, multi-branch, warehouse and business-control foundation. Phase 7 is an incremental upgrade. Inspect the existing repository first. Do not rebuild working functionality, reset the database, or delete/corrupt existing tenant data.

1. Mission

Make AK VisionFlow a specialized end-to-end platform for optical shops, eye-care businesses and combined optical/clinic operations. The core differentiator is the seamless connection between patient care, eye examination, prescription, optical product selection, lab/job processing, billing, payment and customer history.

2. Product Positioning

Patient → Examination → Prescription → Optical Selection → Optical Order → Lab → Fitting → Invoice → Payment → Delivery → Follow-up

The platform must remain generic multi-tenant SaaS. Never hard-code a specific clinic, doctor, shop, terminology or business identity.

3. Non-Negotiable Rules

- Preserve all existing Phase 1–6 functionality.
- Never reset, truncate, recreate or destructively migrate the production database.
- All schema changes must use safe, versioned migrations.
- Existing customers, suppliers, products, inventory, sales, purchases, accounting, branches, warehouses, optical orders, prescriptions and payments must remain intact.
- Maintain strict tenant isolation for patients, doctors, appointments, clinical records, prescriptions and optical records.
- Clinical and patient information must be protected by strong backend authorization and role-based access.
- Do not expose clinical/patient information to users who do not have permission.
- Maintain auditability for sensitive clinical and financial actions.
- Preserve offline/sync safety and idempotency for supported workflows.
- Run existing automated tests and add comprehensive Phase 7 regression tests.

4. Patient Management

- Patient registration and unique patient identifier.
- Patient demographic/contact information.
- Emergency/contact details where appropriate.
- Patient status and active/inactive handling.
- Patient search by name, phone, ID and other permitted fields.
- Patient profile with 360-degree history.
- Previous visits.
- Previous prescriptions.
- Previous optical orders.
- Previous purchases.

- Payments and outstanding balance.
- Clinical notes/history with appropriate access control.
- Attachments/documents where supported by the existing architecture.
- Merge/duplicate-patient detection with safe, controlled handling.

5. Doctor & Provider Management

- Doctor/provider profiles.
- Specialty/designation.
- Branch/location assignment.
- Availability/schedule.
- Appointment capacity.
- Doctor-wise appointments.
- Doctor-wise patient count.
- Doctor-wise clinical/optical activity reports.
- Permission-aware access to clinical information.

6. Appointment & Queue Management

- Appointment booking.
- Appointment calendar.
- Doctor selection.
- Branch selection.
- Appointment types.
- Appointment status: scheduled, confirmed, arrived, in-progress, completed, cancelled, no-show.
- Token/queue management.
- Today's queue dashboard.
- Waiting-time visibility.
- Follow-up appointment scheduling.
- Appointment reminders through the Phase 8 communication layer where available.
- Conflict detection to avoid double-booking.

7. Eye Examination / Refraction

Create a structured, user-friendly eye examination workflow. The exact clinical fields should be configurable enough for different practices while maintaining a consistent core model.

- Visual acuity.
- Refraction.
- Right Eye (OD).
- Left Eye (OS).
- Both Eyes (OU) where applicable.
- Sphere (SPH).
- Cylinder (CYL).

- Axis.
- ADD.
- PD / pupillary distance where applicable.
- Near/Distance values where applicable.
- Other refraction/vision measurements supported by the practice.
- Clinical observations/notes.
- Diagnosis/assessment.
- Treatment/advice.
- Follow-up date.
- Examiner/doctor and timestamp.
- Historical examination comparison.

8. Prescription Management

- Create prescription directly from examination.
- Store prescription version/history rather than overwriting history.
- OD/OS/OU support.
- Distance and near prescription where applicable.
- Lens-related parameters.
- Prescription issue date.
- Prescribing doctor.
- Patient linkage.
- Print/download/share-ready prescription format.
- Link prescription to optical order.
- Show previous prescription for comparison.
- Record corrections/replacements without silently changing historical prescriptions.

9. Optical Product & Selection Workflow

- Frame selection.
- Frame brand/model/size/colour where applicable.
- Lens selection.
- Lens type.
- Index/material.
- Coating.
- Design/type such as single vision/progressive/bifocal where applicable.
- Contact lens selection where supported.
- Accessories and fitting charges.
- Price and discount.
- Prescription-to-product compatibility checks where practical.
- Show customer previous frame/lens purchase history.

10. Advanced Optical Order / Job Card

Prescription → Frame/Lens Selection → Order → Lab → Production/Fitting → Quality Check → Ready → Delivery

- Create job/order from prescription.
- Job card number.
- Customer/patient linkage.
- Frame and lens details.
- Lab selection.
- In-house or third-party lab.
- Lab cost and expected delivery.
- Job status lifecycle.
- Expected completion date.
- Delay flag.
- Quality-control step.
- Fitting details.
- Ready-for-delivery status.
- Customer delivery confirmation.
- Job profitability.
- Full job history and audit trail.

11. Optical Lab Management

- In-house lab work queue.
- Third-party lab work queue.
- Lab-wise pending jobs.
- Jobs by status.
- Delayed jobs.
- Expected delivery tracking.
- Lab cost tracking.
- Lab performance metrics.
- Turnaround-time reporting.
- Rejected/rework tracking.
- Quality-check record.
- Link every job back to patient/customer, prescription and invoice.

12. CRM — Customer 360°

- Unified customer/patient profile.
- Purchase history.
- Prescription history.
- Examination history.
- Optical order history.

- Frame/lens history.
- Payment history.
- Outstanding balance.
- Appointments.
- Follow-ups.
- Complaints/service notes where appropriate.
- Customer segmentation.
- New vs returning customer analysis.
- Inactive customer identification.
- High-value customer identification.
- Customer retention indicators.

13. Clinical-to-Commercial Integration

This is a critical Phase 7 requirement. Avoid isolated modules. The same patient/customer journey should flow through clinical and commercial operations without duplicate data entry.

- Patient registration should be available from appointment, clinic and optical workflows.
- Completed examination can create/update a prescription.
- Prescription can initiate an optical order.
- Optical order can create a job card/lab job.
- Optical order can create a quotation/invoice using the existing financial/POS engine.
- Payment must use the existing payment/accounting infrastructure.
- Delivery must update order status and customer history.
- All events must remain traceable from the patient/customer profile.

14. Clinical & Sensitive Data Security

- Separate clinical permissions from ordinary POS permissions where necessary.
- Restrict patient clinical records to authorized roles.
- Use backend authorization for every clinical endpoint.
- Audit access/change actions for sensitive clinical records.
- Do not include sensitive clinical details in ordinary sales reports unless authorized.
- Avoid exposing clinical data in logs.
- Protect file/document attachments if implemented.
- Ensure tenant isolation for all patient/clinical records.
- Define retention/archival behavior without destructive deletion of required history.

15. Dashboard / Command Center Integration

- Today's appointments.
- Patients seen today.
- Waiting queue.
- Completed examinations.

- Prescriptions issued.
- Optical orders created.
- Pending optical jobs.
- Delayed jobs.
- Ready orders.
- Delivered orders.
- Doctor/branch performance.
- New vs returning patients/customers.
- Outstanding optical payments.
- Conversion from examination → optical order where meaningful.
- Drill-down into source records.
- Respect all branch, tenant and role permissions.

16. Reports

- Patient visit report.
- Appointment report.
- Doctor performance report.
- Examination/refraction report.
- Prescription history report.
- Optical order report.
- Lab job report.
- Pending/delayed job report.
- Turnaround-time report.
- Optical sales/profitability report.
- Customer retention/repeat-purchase report.
- New vs returning customer report.
- Customer outstanding report.
- Branch-wise clinic/optical report.
- Prescription-to-order conversion report.

17. UX Requirements

- Clinical workflows must be fast and keyboard-friendly for receptionists and optometrists.
- Minimize repeated data entry.
- Make OD/OS/OU fields visually clear.
- Use structured forms rather than relying on free text for core clinical measurements.
- Provide quick access to previous prescriptions.
- Make patient search extremely fast.
- Use clear appointment/queue statuses.
- Keep ordinary shop users away from unnecessary clinical complexity.

- Maintain existing responsive/PWA/light-dark design.
- Support printing of prescriptions and relevant optical documents.

18. Offline & Sync Requirements

- Support offline patient/appointment/optical workflows where safe under the existing offline architecture.
- Clinical records that require stronger consistency should have explicit conflict rules.
- Use idempotency for synced patient, appointment, prescription and optical-order operations.
- Prevent duplicate patients, prescriptions, orders and payments after retries.
- Never allow offline synchronization to bypass tenant or role permissions.
- Display unresolved sync/conflict states clearly.

19. Testing Requirements

- Test patient/doctor/appointment tenant isolation.
- Test clinical RBAC and unauthorized access.
- Test patient registration and duplicate handling.
- Test appointment conflict detection.
- Test examination and prescription creation.
- Test prescription history/versioning.
- Test prescription → optical order integration.
- Test optical order → lab job → delivery lifecycle.
- Test financial integration with existing invoice/payment/accounting engine.
- Test branch-level access.
- Test offline retry/idempotency for supported workflows.
- Test sensitive-data audit logging.
- Run the complete regression suite from previous phases.

20. Migration & Data Safety

Inspect the existing customer, prescription and optical-order models before adding patient/clinical relationships. Where a customer already represents a patient, design a safe relationship rather than duplicating identities. Do not rewrite historical prescriptions or optical orders destructively. If migration is required, preserve existing IDs and history and document any assumptions.

21. Required Deliverables

- Patient Management.
- Doctor/Provider Management.
- Appointment and Queue Management.
- Eye Examination/Refraction module.
- Prescription history/versioning.
- Advanced Optical Product Selection.
- Advanced Optical Order/Job Card.
- In-house and third-party Lab Management.

- Customer/Patient 360° CRM.
- Clinical-to-commercial integrated workflow.
- Sensitive clinical data security and audit controls.
- Command Center integration.
- Reports and drill-downs.
- Safe database migrations.
- Automated tests and regression results.
- Updated documentation and Phase 7 completion report.

22. Acceptance Criteria

- Existing Phase 1–6 functionality continues to work.
- No existing data is deleted, reset or corrupted.
- Patients can be registered and searched quickly.
- Appointments and queues work reliably.
- Eye examinations and refraction data are structured and retrievable.
- Prescriptions are versioned and historically preserved.
- Prescription can flow directly into an optical order.
- Optical orders support lab/job tracking through delivery.
- In-house and third-party lab workflows are supported.
- Patient/customer history provides a true 360° view.
- Clinical data is protected by backend RBAC and tenant isolation.
- Financial transactions continue to use the existing authoritative accounting/POS engine.
- Command Center displays accurate clinical/optical KPIs.
- All existing tests plus new Phase 7 tests pass.
- No critical security, privacy or data-integrity issue remains open.

23. Claude Execution Order

1. Inspect the current repository, database schema and existing customer/prescription/optical modules.
2. Run the complete existing test suite and establish a baseline.
3. Design the patient/clinical data model and relationships with existing customers and optical records.
4. Design and test safe migrations.
5. Implement patient and doctor management.
6. Implement appointments and queue.
7. Implement examination/refraction workflow.
8. Implement prescription versioning/history.
9. Implement advanced optical selection and job-card workflow.
10. Implement lab tracking and delivery lifecycle.
11. Implement Customer/Patient 360° CRM.
12. Integrate clinical and commercial workflows.

13. Integrate Command Center and reports.
14. Implement security/audit controls.
15. Add comprehensive automated tests.
16. Run full regression, security and data-integrity review.
17. Document implementation, migrations, limitations and operational procedures.

Final Instruction to Claude

Build Phase 7 as a safe, production-quality incremental upgrade. Do not rebuild the existing core and do not reset the database. The goal is to make AK VisionFlow a genuinely integrated Optical + Eye-Care platform, not a collection of disconnected modules. Patient, examination, prescription, optical order, laboratory, billing, payment and follow-up workflows must connect naturally. Clinical data must have stronger privacy and role-based access than ordinary business records. Preserve existing accounting, inventory, multi-branch, offline/sync and Command Center functionality. Do not introduce a heavy hospital-management system in this phase; focus on high-value eye-care, optical and CRM workflows that fit the target market. The result must be tested, maintainable and ready for Phase 8: WhatsApp + Automation + Customer Portal.